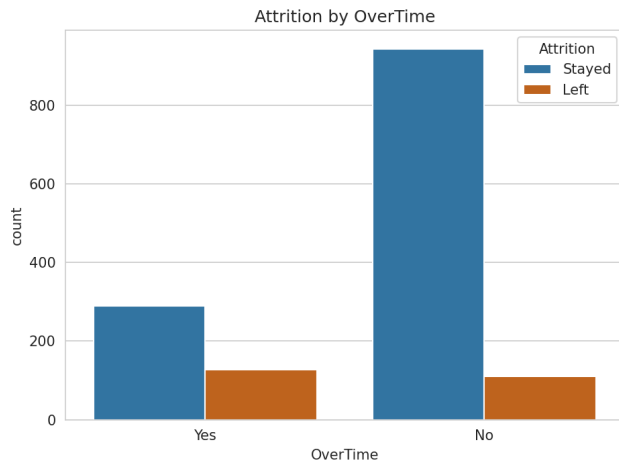


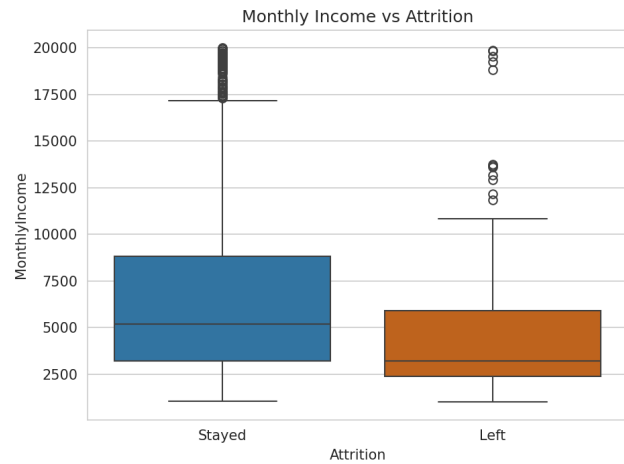
HR Employee Attrition Analysis

What actually predicts attrition in the IBM HR dataset, and what to do about it

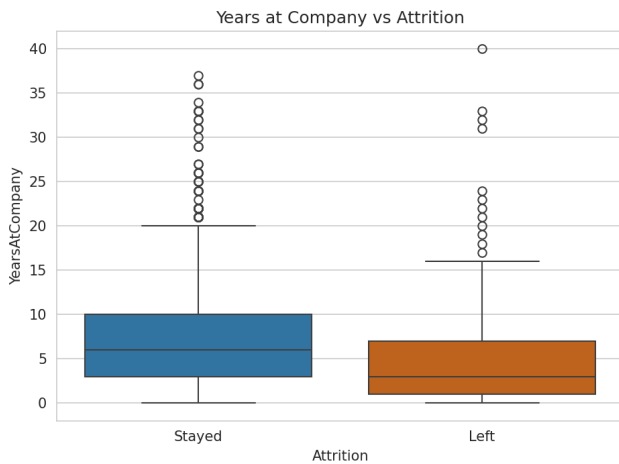
1,470 employees | 35 features | Logistic Regression + Random Forest, class-weight balanced



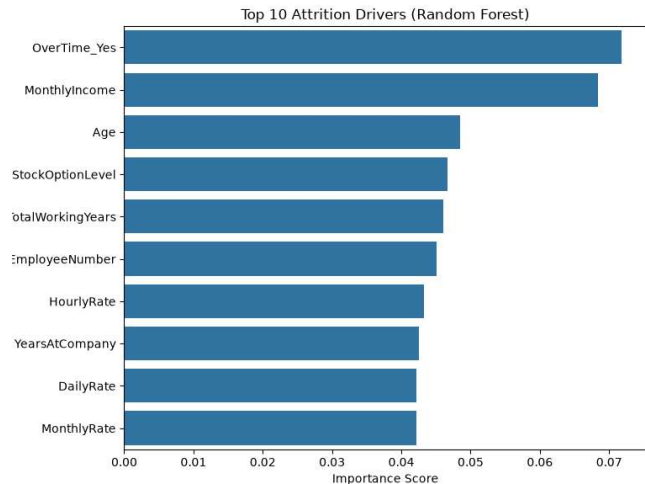
Employees working overtime leave at ~31%, vs. 10% for everyone else -- roughly 3x higher, and the top feature in both the correlation heatmap and Random Forest importances.



Income is a real factor (median Rs 3,100 for leavers vs. Rs 5,200 for stayers) but ranks second -- not the main story people assume it is.



Median tenure for leavers is ~3 years vs. ~6 for those who stayed -- this reads as an early-career retention gap, not a company-wide problem.



OverTime and MonthlyIncome are the top two drivers in the Random Forest's own feature importances -- confirming the same story as the manual EDA and correlation heatmap, not just one method agreeing with itself

What HR should do with this

- Address overtime load directly -- it's the single biggest lever, not a side factor.
- Focus retention efforts on the 0-3 year tenure window, where the actual risk concentrates.
- Use Logistic Regression (or its recall) for flagging at-risk employees, not the highest-accuracy model.